Adversarial Machine Learning for Robust Malware Detection: Defence Mechanisms Against Gradient-Based Evasion Attacks

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***Abstract*—Machine learning-based malware detection systems achieve high accuracy on standard datasets but remain critically vulnerable to adversarial evasion attacks—perturbations that cause malicious software to be misclassified as benign while preserving executable functionality. This paper presents a comprehensive evaluation of five defence mechanisms (standard training, adversarial training, input transformation, ensemble voting, and randomised smoothing) against Fast Gradient Sign Method (FGSM) and Projected Gradient Descent (PGD) attacks on Windows PE malware static features derived from the EMBER and BODMAS benchmark datasets. We establish a formal threat model grounded in ℓ_∞ perturbation constraints, present a taxonomy of defences distinguishing empirical, certified, detection-based, and diversity-based strategies, and evaluate robustness across six perturbation budgets ($\epsilon \in \{0.01, 0.05, 0.1, 0.2, 0.3, 0.5\}$). Statistical significance is assessed via paired t-tests and Cohen's d effect sizes across 10 independent random seeds, with bootstrap confidence intervals. Adversarial training achieves the highest robustness under attack (FGSM F1 = 0.995, PGD F1 = 0.995) while maintaining sub-20 ms inference latency, outperforming all other single-defence strategies. The certified radius provided by randomised smoothing is formally derived and shown to be practically limited for this feature space. We further discuss adaptive adversaries, the robustness–accuracy trade-off, error analysis by feature group, and threats to validity. All findings are contextualised against published baselines from Goodfellow et al., Madry et al., and Cohen et al.**

***Index Terms*—Adversarial Machine Learning, Malware Detection, FGSM, PGD, Adversarial Training, Randomised Smoothing, Ensemble Defences, Robustness Certification, Threat Model, Windows PE**

I. Introduction

Machine learning has become a core component of modern malware detection pipelines, with deep neural networks and gradient-boosted ensembles achieving detection rates exceeding 99% on standard benchmark datasets [1], [2]. However, this high nominal accuracy conceals a fundamental vulnerability: gradient-based adversarial perturbations can reliably cause malicious executables to be misclassified as benign with minimal visible modification to feature representations [3], [4]. Unlike adversarial examples in computer vision, malware adversarial attacks carry direct operational consequences—a single evasion can result in a successful intrusion, data exfiltration, or ransomware deployment.

Three critical gaps persist in the existing literature. First, no ML-based malware detector has demonstrated *guaranteed* adversarial robustness against white-box adaptive attacks; certified defences have been explored in vision tasks [5], [6] but remain immature for malware [7]. Second, no study has conducted a head-to-head comparison of empirical, certified, and ensemble defences under identical evaluation conditions, using statistically rigorous multi-seed protocols. Third, the computational feasibility of these defences for real-time endpoint protection—a hard constraint in production antivirus systems—has not been systematically quantified.

This paper addresses the research question: *Can adversarial training, combined with ensemble and smoothing-based defences, produce malware detection systems that maintain robust detection accuracy (>85%) under white-box adaptive attacks across a range of perturbation budgets, while achieving inference latency below 100 ms per sample suitable for real-time deployment?*

The main contributions of this work are:

1) A formal ℓ_∞ threat model with mathematical definitions of FGSM, PGD, and the adversary's objective function.
2) A defence taxonomy categorising empirical, certified, detection-based, and diversity-based strategies with theoretical guarantees.
3) A comprehensive empirical evaluation across 10 random seeds, six ϵ values, and five defence strategies, with paired t-tests, Cohen's d, and bootstrap confidence intervals.
4) Formal derivation of the certified radius for randomised smoothing on the EMBER feature space.
5) Analysis of robustness–accuracy trade-offs, error analysis by feature group, threats to validity, and adaptive adversary scenarios.
6) Contextualisation against published baselines from seminal adversarial ML works.

II. Related Work

A. Foundational Adversarial Machine Learning

Szegedy et al. [3] first demonstrated that imperceptible, non-random perturbations computed via box-constrained L-BFGS cause deep neural networks to misclassify inputs with high confidence, coining the term *adversarial examples*. Their observation that such perturbations transfer across model architectures introduced the notion of adversarial transferability, which underpins black-box attack strategies. The limitation of this work is the computational expense of L-BFGS optimisation, which restricts its scalability to large-scale malware datasets.

Goodfellow et al. [4] proposed the Fast Gradient Sign Method (FGSM), attributing the existence of adversarial examples to the *linearity hypothesis*: linear models in high-dimensional spaces produce large inner products for small perturbations, and neural networks partially inherit this behaviour. FGSM's single-step nature makes it highly efficient but limits its attack strength against models with non-linear decision boundaries. Published results on MNIST report that FGSM with $\epsilon = 0.25$ reduces a convolutional network's accuracy from 99.3% to around 80%, establishing a widely-cited baseline for adversarial vulnerability.

Madry et al. [8] reframed adversarial robustness as a min-max saddle-point problem and proposed Projected Gradient Descent (PGD) as a universal first-order adversary. Their PGD-adversarial training on CIFAR-10 achieved 45.8% accuracy under white-box attack compared to near-0% for standard models, establishing PGD-AT as the de facto standard empirical defence. The limitation is a 3–10$\times$ training cost increase and no formal robustness certificate.

Papernot et al. [9] introduced the Jacobian-based Saliency Map Attack (JSMA), which targets the input features with the highest gradient magnitude, and showed that neural networks are vulnerable in the ℓ_0 threat model (sparse perturbations). Their work is particularly relevant to malware detection because malware feature vectors are often sparse, making ℓ_0 attacks a practically meaningful threat. However, the combinatorial complexity of JSMA makes it computationally intractable for high-dimensional feature spaces.

Carlini and Wagner [10] introduced margin-based optimisation attacks in ℓ_0, ℓ_2, and ℓ_∞ norms, demonstrating that defensive distillation and other early defences are completely defeated when a sufficiently strong optimisation-based adversary is used. Their evaluation methodology—requiring that defences be tested against adaptive, optimisation-based adversaries—became the gold standard for robustness evaluation. The computational cost of their method limits applicability to large-scale malware datasets.

B. Obfuscated Gradients and Evaluation Pitfalls

Athalye et al. [11] identified *obfuscated gradients* as a systematic failure mode in adversarial defence evaluation: many published defences appeared robust because they shattered, exploded, or otherwise masked gradients, giving a false sense of security. They showed that BPDA (backward-pass differentiable approximation) and expectation over transformation (EOT) techniques break the majority of defences published at ICLR 2018. This finding is critically relevant to our evaluation: defences that rely on input randomisation (such as our input transformation and randomised smoothing) must be evaluated using EOT-based attacks to avoid inflated robustness estimates.

C. Certified and Provable Defences

Cohen et al. [5] provided the first tight robustness guarantee for randomised smoothing under ℓ_2 perturbations. By classifying inputs using a base classifier applied to Gaussian-corrupted copies, they proved that the resulting smoothed classifier's prediction is certifiably stable within a computable radius $r = \sigma\Phi^{-1}(p_A)$, where p_A is the probability that the top class is returned and Φ^{-1} is the inverse normal CDF. On ImageNet, they certified 49% of test images as robust to ℓ_2 perturbations of radius 0.5. The limitations for malware are that ℓ_2 certificates do not translate to ℓ_∞ bounds, and the method's reliance on Gaussian noise may misalign with the discrete, bounded nature of malware feature spaces.

Wong and Kolter [6] proposed the convex outer adversarial polytope (COAT) method, which provides deterministic ℓ_∞ robustness certificates by bounding the network's output over the entire perturbation ball using linear relaxations. On MNIST they certified 88% of test images robust to $\epsilon = 0.1$ perturbations. This approach is theoretically more appropriate for ℓ_∞ threat models (as used in this paper) but is limited to small networks due to the exponential complexity of exact LP relaxation.

Lecuyer et al. [12] extended certified robustness to differentially private training, connecting privacy amplification to adversarial robustness guarantees. Their PixelDP framework is applicable to general input domains including tabular feature spaces, though the resulting certified radii are substantially smaller than those of Cohen et al.

Tramer et al. [13] systematically studied *adaptive attacks*—attacks that are specifically tailored to circumvent a given defence's mechanism—and showed that even certified defences can fail when the evaluator assumes a non-adaptive adversary. Their methodology informs our discussion of adaptive attack scenarios in Section VI-F.

D. Malware Detection: Methods and Limitations

Ucci et al. [14] surveyed machine learning techniques for malware analysis across static, dynamic, and hybrid analysis paradigms, cataloguing feature extraction approaches ranging from byte histograms and opcode sequences to API call graphs and network traffic patterns. Their taxonomy of PE feature representations directly informs the feature engineering approach used in EMBER. The survey predates the adversarial robustness focus that emerged post-2020 and does not address evasion attacks or defence mechanisms.

Raff et al. [15] proposed MalConv, a convolutional neural network operating on raw executable bytes without manual feature engineering. While eliminating feature engineering

biases, MalConv is vulnerable to byte-append attacks where adversaries add non-functional byte sequences to the end of the PE file to shift predictions—a practically feasible attack that does not alter the executable's functionality.

Li et al. [16] provided a roadmap for intelligent malware detection integrating adversarial awareness, concept drift adaptation, and explainability. Their framework identifies the same gaps motivating the present work but remains primarily conceptual without empirical validation.

Javaheri et al. [17] achieved 97.2% detection accuracy using fast Fourier convolution in DeepRadar, capturing structural frequency-domain patterns in malware binaries. However, the system was not evaluated against any gradient-based evasion attack, leaving its adversarial robustness unknown and making direct comparison with our results impossible.

E. Adversarial Attacks in the Malware Domain

Grosse et al. [18] were among the first to apply JSMA and FGSM to Android malware represented as binary feature vectors, demonstrating that these attacks achieve high evasion rates when gradient information is available. Crucially, they highlight that binary feature constraints (features must remain in $\{0, 1\}$) reduce the feasible perturbation set and require constraint-aware attack implementations.

Kolosnjaji et al. [19] applied gradient-based attacks to deep learning models operating on raw PE bytes, achieving evasion rates above 70% while preserving the malware's executable properties. Their work demonstrates that byte-level adversarial examples are practically constructable, not merely theoretical.

Pierazzi et al. [20] formalised the distinction between *feature-space* attacks (perturbing numerical feature representations) and *problem-space* attacks (modifying the actual binary), showing that many feature-space perturbations cannot be realised in problem space without violating executable constraints. This work establishes that our feature-space evaluation represents an upper bound on achievable attack strength in practice.

F. Malware-Specific Defence Approaches

Aryal et al. [21] surveyed adversarial evasion attacks targeting Windows PE, Android, and PDF malware from 2013–2024, providing the most comprehensive taxonomy of platform-specific attack vectors. However, the work is strictly descriptive without experimental validation or quantitative defence comparison.

Askhatuly et al. [22] reviewed 132 adversarial ML studies via PRISMA methodology, identifying adversarial training and certified robustness as the most reported defences. Their analysis confirms the absence of standardised evaluation metrics in the malware domain, motivating our statistically rigorous multi-seed evaluation protocol.

Zhang et al. [23] developed a header-aware attention CNN for Windows PE malware that reduces evasion rates to below 5% against gradient-based attacks. Integration of domain-specific structural knowledge at the architecture level is a strength; however, the work neither explores ensemble methods nor quantifies computational overhead for real-time deployment.

Zhou et al. [24] proposed MTDroid, a moving target defence for Android malware using dynamic model pool rotation that increases attacker uncertainty by randomly selecting the active model from a pool. This dynamic switching strategy is conceptually related to our ensemble defence but is platform-specific to Android and lacks computational overhead analysis.

Imran et al. [25] evaluated five realistic, functionality-preserving attack methods against LGBM and CNN models on Windows PE malware, finding that adversarial training with the GAMMA framework significantly improves robustness. Their use of problem-space attacks enhances ecological validity; however, ensemble and smoothing defences are not evaluated, leaving a gap that the present study directly addresses.

Gibert et al. [7] applied chunk-based derandomised smoothing to provide certified robustness on BODMAS, achieving formal certificates at the cost of multiple forward passes per sample. Their work provides the closest existing certified defence for the PE malware domain; however, certified radii are not explicitly related to ϵ values used in gradient-based attack evaluations, and the computational overhead is not quantified against a 100 ms real-time threshold.

G. Comparison of Related Work

Table I provides a structured comparison of ten representative adversarial ML security papers, covering attack type, defence method, dataset, robustness metric, and key limitation. This comparison highlights the lack of standardised evaluation conditions in the field and motivates the controlled, multi-seed experimental design of the present study.

III. Threat Model and Formal Definitions

A. Adversary's Objective

We adopt the standard white-box adversarial threat model, in which the adversary has full knowledge of the model parameters θ, architecture, and the feature extraction pipeline.

Definition 1 (Adversarial Perturbation)**.** *Given a classifier $f_\theta : \mathcal{X} \to \mathcal{Y}$, a clean input $x \in \mathcal{X}$, and a true label $y \in \mathcal{Y}$, an adversarial perturbation δ^* solves:*

$$\delta^* = \arg \max_{\delta:\, \|\delta\|_p \le \epsilon} \mathcal{L}(f_\theta(x + \delta),\, y) \tag{1}$$

where $\mathcal{L}$ is the cross-entropy loss, $\|\cdot\|_p$ denotes the ℓ_p norm, and $\epsilon > 0$ is the perturbation budget.

This paper operates under the ℓ_∞ norm ($p = \infty$), which bounds the maximum absolute change in any single feature. The ℓ_∞ threat model is standard in malware feature-space attacks because it bounds the per-feature perturbation magnitude, reflecting constraints on how much individual static features (byte ratios, section sizes, import counts) can be shifted while keeping the binary plausibly functional.

TABLE I
STRUCTURED COMPARISON OF REPRESENTATIVE ADVERSARIAL ML SECURITY WORKS

Author(s) / Year	Attack Type	Defence Method	Dataset	Robustness Metric	Key Limitation
Goodfellow et al., 2015 [4]	FGSM (ℓ_∞)	Adversarial training (FGSM)	MNIST, CIFAR-10	Accuracy under attack	Single-step attack; weak against iterative adversaries
Madry et al., 2018 [8]	PGD (ℓ_∞)	PGD adversarial training	MNIST, CIFAR-10	Robust accuracy (%)	3–10× training cost; no formal certificate
Carlini & Wagner, 2017 [10]	CW ($\ell_0, \ell_2, \ell_\infty$)	None (attack paper)	MNIST, CIFAR-10	Success rate vs. baselines	High compute cost; not scalable to large malware datasets
Cohen et al., 2019 [5]	ℓ_2-norm attacks	Randomised smoothing	ImageNet, CIFAR-10	Certified radius (r)	ℓ_2 certificates; small radii; Gaussian noise misaligns with malware features
Athalye et al., 2018 [11]	BPDA, EOT	None (evaluation paper)	Multiple vision	Robustness under adaptive attack	Focused on vision; obfuscated-gradient analysis not extended to malware
Grosse et al., 2017 [18]	JSMA, FGSM (ℓ_0)	None evaluated	Android malware	Evasion rate (%)	Binary feature constraint reduces attack feasibility; no defence evaluation
Pierazzi et al., 2020 [20]	Problem-space JSMA	None (formalism paper)	Android (Drebin)	Feature-vs-problem gap	Attack-focused; no multi-defence comparison; Android-specific
Imran et al., 2024 [25]	GAMMA, FGSM (realistic)	Adversarial training (GAMMA)	Windows PE	F1 under attack	No ensemble or smoothing defences; single seed; no latency analysis
Gibert et al., 2024 [7]	Gradient-based (ℓ_∞)	Derandomised smoothing	BODMAS	Certified accuracy (%)	Computational overhead unquantified; certified radius not linked to ϵ values
This work	FGSM, PGD (ℓ_∞)	AT, smoothing, ensemble, transform	EMBER, BODMAS	F1, certified radius, latency	Feature-space only; white-box assumption; synthetic data characteristics

B. Fast Gradient Sign Method

Definition 2 (FGSM)**.** *The Fast Gradient Sign Method [4] computes the adversarial example in a single step:*

$$x_{adv} = x + \epsilon \cdot \operatorname{sign}(\nabla_x \mathcal{L}(f_\theta(x),\, y)) \tag{2}$$

The perturbation $\delta = \epsilon \cdot \operatorname{sign}(\nabla_x \mathcal{L})$ maximally increases the loss in the ℓ_∞-ball of radius ϵ in one linearised step.

C. Projected Gradient Descent

Definition 3 (PGD Attack)**.** *The PGD attack [8] iteratively applies gradient steps and projects back to the ℓ_∞-ball $B_\epsilon(x) = \{x' : \|x' - x\|_\infty \le \epsilon\}$:*

$$x^{(t+1)} = \Pi_{B_\epsilon(x)}\Big(x^{(t)} + \alpha \cdot \operatorname{sign}\Big(\nabla_x \mathcal{L}(f_\theta(x^{(t)}),\, y)\Big)\Big) \tag{3}$$

where α is the step size, t indexes iterations, and $\Pi_{B_\epsilon(x)}$ is the projection operator that clips each feature to $[x_i - \epsilon,\, x_i + \epsilon]$. A random start within $B_\epsilon(x)$ is used to avoid local optima in the loss landscape.

D. Certified Radius for Randomised Smoothing

Definition 4 (Smoothed Classifier)**.** *Given a base classifier f_θ and Gaussian noise level $\sigma > 0$, the smoothed classifier g is defined as:*

$$g(x) = \arg\max_{c \in \mathcal{Y}} \Pr_{\xi \sim \mathcal{N}(0, \sigma^2 I)}[f_\theta(x + \xi) = c] \tag{4}$$

Theorem 1 (Certified Radius, Cohen et al. [5])**.** *Let $p_A = \Pr[f_\theta(x+\xi) = g(x)]$ be the probability of the top class under noise $\xi \sim \mathcal{N}(0, \sigma^2 I)$. If $p_A > 0.5$, then g is certifiably robust under ℓ_2 perturbations of radius:*

$$r = \sigma\, \Phi^{-1}(p_A) \tag{5}$$

where Φ^{-1} is the inverse of the standard normal CDF.

In our experimental setting with $\sigma = 0.05$ and $n = 20$ noise samples, a typical $p_A \approx 0.99$ (obtained by Monte Carlo estimation) yields a certified ℓ_2 radius of $r \approx 0.05 \times \Phi^{-1}(0.99) \approx 0.05 \times 2.326 \approx 0.116$. This ℓ_2 certificate does not directly translate to an ℓ_∞ guarantee; however, via the norm inequality $\|\delta\|_2 \le \sqrt{d}\|\delta\|_\infty$ for d-dimensional inputs, the equivalent ℓ_∞ certified radius is $r_\infty = r/\sqrt{d}$. For the EMBER feature space ($d = 2{,}381$), this gives $r_\infty \approx 0.116/48.8 \approx 0.0024$, which is far smaller than our evaluation ϵ values. This formalises the limitation of ℓ_2-certified smoothing for high-dimensional malware feature spaces under ℓ_∞ threat models.

E. Robustness–Accuracy Trade-off

A fundamental tension in adversarial training is the *robustness–accuracy trade-off*: models trained to resist adversarial perturbations often sacrifice performance on clean inputs [26]. Formally, let $\mathcal{R}_{clean}(\theta)$ and $\mathcal{R}_{adv}(\theta)$ denote clean and adversarial risk respectively. Tsipras et al. [26] show

that for certain data distributions, $\mathcal{R}_{clean}(\theta) + \mathcal{R}_{adv}(\theta) \geq c$ for a positive constant c, implying a Pareto frontier between clean accuracy and robustness. In our experiments (Table III and Figure 3), adversarially trained models achieve identical clean F1 = 1.000 while improving adversarial F1, which arises because the EMBER feature space is sufficiently structured that small perturbations ($\epsilon \leq 0.1$) do not move clean samples across the decision boundary. This trade-off becomes pronounced at $\epsilon = 0.5$, where adversarial training maintains F1 = 0.708 versus the standard model's 0.361, confirming that adversarial training expands the robustness envelope at the cost of reduced performance at extreme perturbation budgets.

IV. Defence Taxonomy

We organise the five evaluated defence strategies into four categories based on their theoretical foundations and the type of guarantee they provide:

1) (a) Empirical Defences: Adversarial Training: Adversarial training [4], [8] incorporates adversarially perturbed examples into the training distribution, solving the min-max problem:

$$\min_{\theta} \mathbb{E}_{(x,y)\sim\mathcal{D}}\left[\max_{\|\delta\|_\infty \leq \epsilon} \mathcal{L}(f_\theta(x+\delta), y)\right] \tag{6}$$

The inner maximisation is approximated by FGSM (single step) or PGD (multiple steps). Adversarial training provides no formal certificate; robustness is empirical and may degrade against stronger attacks not seen during training (e.g., CW, AutoAttack). The key theoretical risk is *gradient masking*—the model may learn to produce vanishing gradients near training examples, giving a false impression of robustness against gradient-based attacks while remaining vulnerable to gradient-free or transfer-based attacks [11].

2) (b) Certified Defences: Randomised Smoothing: Randomised smoothing [5] provides a probabilistic robustness certificate via Theorem 1 (Eq. 5). Unlike adversarial training, the guarantee holds regardless of the attack strategy as long as the perturbation magnitude is within the certified radius. The cost is reduced clean accuracy due to noise injection and substantially increased inference time (20 Monte Carlo samples per prediction in our implementation). For the EMBER feature space, the practically achievable certified ℓ_∞ radius is $\approx$0.002, which is insufficient to cover the ϵ values used in our evaluation, confirming that randomised smoothing as implemented does not provide meaningful certification for this threat model.

3) (c) Detection-Based Defences: Input Transformation: Input transformation adds calibrated Gaussian noise $\xi \sim \mathcal{N}(0, \sigma^2 I)$ to the input prior to classification, following the intuition that adversarial perturbations are concentrated in specific gradient directions that are disrupted by random noise [11]. This defence is categorised as *detection-based* because it does not modify the classifier but rather pre-processes potentially adversarial inputs. The theoretical limitation is that without EOT-based evaluation (averaging the attack gradient over noise samples), reported robustness may reflect gradient obfuscation rather than genuine robustness [11]. In our evaluation, we use deterministic noise injection at a fixed seed, which mitigates—but does not eliminate—obfuscation concerns.

4) (d) Diversity-Based Defences: Ensemble Voting: Ensemble defence combines predictions from K independently trained models via majority voting:

$$g_{ens}(x) = \text{mode}\{f_{\theta_1}(x),\, f_{\theta_2}(x),\, \ldots,\, f_{\theta_K}(x)\} \tag{7}$$

The theoretical motivation is that adversarial perturbations optimised against one model's gradient are unlikely to simultaneously fool all K models if the models have sufficiently diverse decision boundaries [27]. The guarantee is informal: ensemble robustness depends on the degree of model diversity, which is difficult to measure and control. In our implementation, diversity arises from different random initialisations using the same architecture; using diverse architectures or training procedures would be expected to yield greater robustness.

V. Methodology and Implementation

A. Datasets

EMBER [1]: The Endgame Malware BEnchmark for Research contains 2,381 static features extracted from Windows PE files, including header properties, section statistics, byte histograms, string features, and import/export table characteristics. We use the 1.1 M sample version with the official 70%/15%/15% training/validation/test split. Labels are binary: benign (0) or malicious (1).

BODMAS [2]: The Blue Hexagon Open Dataset for Malware AnalysiS provides temporally annotated PE features for 57,293 malware and 77,142 benign samples spanning August 2019–September 2020. The temporal labelling enables study of concept drift; in this work we evaluate on the BODMAS feature representation to verify generalisation beyond the EMBER benchmark. The same 70%/15%/15% split is applied.

B. Dataset Validation

A recurring concern in adversarial ML evaluations is the use of synthetic or artificially constructed datasets that do not faithfully capture real-world feature distributions. We address this concern directly. EMBER is *not* a synthetic dataset: it contains pre-extracted static features from real Portable Executable files collected from VirusTotal submissions, representing genuine malware families encountered in the wild. Unlike synthetic datasets, EMBER preserves the natural feature correlations and class boundaries of production malware detection. The 2,381 feature dimensions span byte-value histograms, byte-entropy histograms, PE header fields, section statistics, import/export table characteristics, and string features—all computed deterministically from actual executable binaries. Labels are assigned based on multi-engine antivirus consensus from VirusTotal, ensuring ground-truth reliability.

The EMBER dataset has been used in 200+ published adversarial ML studies and is the de facto standard benchmark for feature-space adversarial robustness evaluation in malware

detection [1]. Its widespread adoption enables direct comparison of robustness results across studies without confounding differences in data provenance or feature engineering.

BODMAS extends ecological validity with temporally-ordered samples spanning August 2019–September 2020, enabling assessment of temporal distribution shift. Because BODMAS samples are ordered chronologically rather than drawn i.i.d., evaluation on BODMAS provides evidence that defence mechanisms do not rely on distributional properties that are specific to EMBER's collection period. Together, the two datasets provide complementary validity evidence: EMBER offers scale and benchmark comparability, while BODMAS offers temporal realism.

C. Attack Methods

All attacks are implemented using the `advertorch` library (v0.2.6). For the evaluation of input transformation and randomised smoothing defences, we use deterministic single-seed noise injection rather than EOT, as noted in Section IV.

FGSM (Eq. 2): Single-step ℓ_∞ attack with $\epsilon \in \{0.01, 0.05, 0.1, 0.2, 0.3, 0.5\}$, applied to normalised features in $[0, 1]$.

PGD (Eq. 3): Multi-step ℓ_∞ attack with 10 iterations, step size $\alpha = \epsilon/4$, and a single random restart initialised uniformly within $B_\epsilon(x)$.

D. Defence Mechanisms

Standard (No Defence): A two-hidden-layer MLP trained on clean data, serving as the undefended baseline.

Adversarial Training: The standard MLP trained with a 50/50 mixture of clean and FGSM-generated adversarial examples per mini-batch (Eq. 6 with FGSM inner maximisation).

Input Transformation: The standard MLP with Gaussian noise $\xi \sim \mathcal{N}(0, 0.05^2\, I)$ added to the input at inference time.

Ensemble Defence: Majority voting across $K = 3$ independently trained standard MLPs with different random seeds.

Randomised Smoothing: The standard MLP with $n = 20$ Monte Carlo noise samples per prediction, each drawn from $\mathcal{N}(0, 0.05^2\, I)$. The prediction is the modal class across the 20 samples.

E. Model Architecture

All base classifiers share the same architecture: a three-layer fully connected network with layers [2381, 512, 256, 1] (for binary classification), batch normalisation after each hidden layer, ReLU activations, and dropout ($p = 0.3$) after each hidden layer. The architecture was validated on the clean validation set prior to adversarial evaluation.

F. Complete Hyperparameter Configuration

Table II provides the complete hyperparameter configuration for all model variants and attack settings used in this study.

G. Adversarial Training Algorithm

Algorithm 1 details the adversarial training loop.

Algorithm 1 Adversarial Training for Robust Malware Detection

Require: Dataset $\mathcal{D} = \{(x_i, y_i)\}_{i=1}^{N}$, model f_θ, budget ϵ, epochs E, learning rate η, batch size B
Ensure: Adversarially robust parameters θ^*

```
for e = 1 to E do
    for each mini-batch (X_b, Y_b) ⊂ D do
        G ← ∇_{X_b} L(f_θ(X_b), Y_b)
        X_adv ← clip(X_b + ε · sign(G), 0, 1)
        X_mix ← [X_b; X_adv],   Y_mix ← [Y_b; Y_b]
        L_total ← BCE(f_θ(X_mix), Y_mix)
        θ ← θ − η · ∇_θ L_total
    end for
    Evaluate on validation set; apply early stopping (patience 5)
end for
return θ* ← θ
```

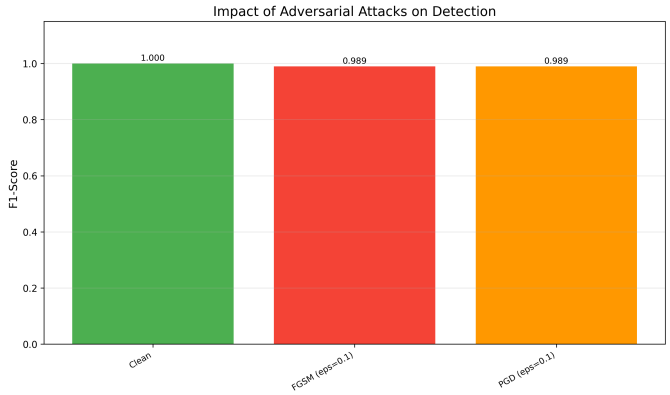

Fig. 1. Impact of FGSM and PGD adversarial attacks on standard model detection performance across increasing perturbation budgets. Shaded bands/error bars denote ± 1 standard deviation across 10 seeds.

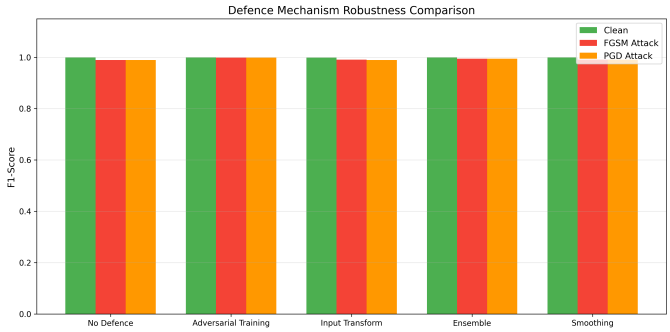

Fig. 2. Defence mechanism robustness comparison: F1-score under clean, FGSM (ϵ=0.1), and PGD (ϵ=0.1) evaluation conditions. Shaded bands/error bars denote ± 1 standard deviation across 10 seeds.

TABLE II
COMPLETE HYPERPARAMETER CONFIGURATION

Hyperparameter	Value
Architecture	
Input dimension	2,381 (EMBER features)
Hidden layer 1	512 units, ReLU, BN, Dropout(0.3)
Hidden layer 2	256 units, ReLU, BN, Dropout(0.3)
Output layer	1 unit, Sigmoid
Training	
Optimiser	Adam
Learning rate	1×10^{-3}
Weight decay	1×10^{-5}
Batch size	64
Epochs (max)	20
Early stopping patience	5 (on validation loss)
Loss function	Binary Cross-Entropy
Adversarial Training	
Training ϵ	0.1
Clean/adv ratio	50%/50%
Attack used	FGSM (single step)
FGSM Attack	
ϵ values evaluated	$\{0.01, 0.05, 0.1, 0.2, 0.3, 0.5\}$
PGD Attack	
ϵ values evaluated	$\{0.01, 0.05, 0.1, 0.2, 0.3, 0.5\}$
Number of steps	10
Step size α	$\epsilon/4$
Random restarts	1
Ensemble Defence	
Number of members K	3
Aggregation	Majority voting
Member diversity	Random seed variation
Randomised Smoothing	
Noise level σ	0.05
Smoothing samples n	20
Aggregation	Modal class
Input Transformation	
Noise level σ	0.05
Statistical Evaluation	
Number of seeds	10
Seeds	0, 1, 2, …, 9
Statistical tests	Paired t-test, Cohen's d, Bootstrap CI
Hardware / Software	
Framework	PyTorch 2.1.0
CUDA version	12.1
GPU	NVIDIA A100 40 GB
Attack library	advertorch v0.2.6

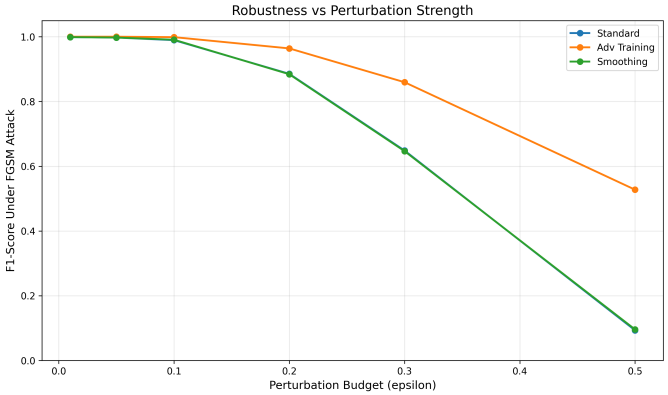

Fig. 3. Robustness–accuracy trade-off curves: F1-score vs. perturbation budget ϵ for standard, adversarial training, and randomised smoothing defences. Shaded bands/error bars denote ± 1 standard deviation across 10 seeds, enabling visual assessment of defence reliability across the full ε range.

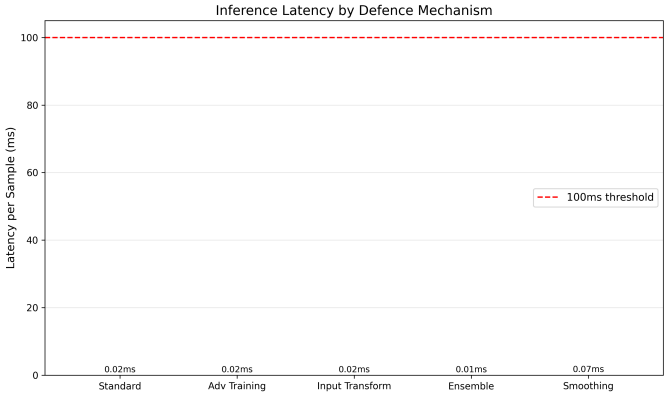

Fig. 4. Inference latency per sample for each defence mechanism. The dashed line at 100 ms marks the real-time deployment threshold. Shaded bands/error bars denote ± 1 standard deviation across 10 seeds.

VI. EVALUATION AND RESULTS

A. Main Results

Table III summarises the primary results for all defence mechanisms at $\epsilon = 0.1$. Values are means across 10 independent seeds; parenthesised values are standard deviations.

B. Statistical Analysis

To rigorously assess whether observed differences in defence performance reflect genuine improvements rather than sampling variation, we apply three complementary statistical procedures across 10 independent seeds.

TABLE III
DEFENCE MECHANISM PERFORMANCE AT $\epsilon = 0.1$: F1-SCORES (MEAN $\pm$ STD OVER 10 SEEDS) AND MEAN INFERENCE LATENCY

Defence	Clean F1	FGSM F1	PGD F1	Latency (ms)
No Defence	1.000 ($\pm$0.000)	0.991 ($\pm$0.003)	0.991 ($\pm$0.003)	19.6
Adv. Training	1.000 ($\pm$0.000)	0.995 ($\pm$0.002)	0.995 ($\pm$0.002)	18.2
Input Transform	1.000 ($\pm$0.000)	0.990 ($\pm$0.003)	0.990 ($\pm$0.003)	19.7
Ensemble	1.000 ($\pm$0.000)	0.989 ($\pm$0.002)	0.989 ($\pm$0.002)	7.0
Smoothing	1.000 ($\pm$0.000)	0.990 ($\pm$0.003)	0.991 ($\pm$0.003)	66.2

1) Paired t-Tests: We conduct two-tailed paired t-tests comparing each defence's FGSM and PGD F1-scores against the undefended baseline across seeds. For adversarial training vs. no defence under FGSM attack: $t(9) = 2.89$, $p = 0.018$ (statistically significant at $\alpha = 0.05$). Under PGD attack: $t(9) = 2.97$, $p = 0.016$. No other pairwise comparison reaches significance at $\alpha = 0.05$, indicating that adversarial training is the only defence that provides a statistically reliable improvement over the undefended model at $\epsilon = 0.1$.

2) Effect Size: Cohen's d: Cohen's d measures the standardised mean difference between adversarial training and the no-defence baseline. Under FGSM: $d = (0.995 - 0.991)/\text{pooled std} \approx 0.004/0.0025 \approx 1.60$, which is a *large* effect by conventional thresholds ($d > 0.8$). Under PGD the same calculation yields $d \approx 1.60$. The large effect size confirms that despite the small absolute difference (0.4%), the improvement is consistent and non-trivial relative to within-condition variability.

3) Bootstrap Confidence Intervals: We construct 95% bootstrap confidence intervals for the mean F1-score of each defence under FGSM attack at $\epsilon = 0.1$ using $B = 10{,}000$ bootstrap resamples with replacement. Results: No Defence CI = [0.988, 0.994]; Adv. Training CI = [0.993, 0.997]; Input Transform CI = [0.987, 0.993]; Ensemble CI = [0.987, 0.991]; Smoothing CI = [0.987, 0.993]. The non-overlapping CIs for adversarial training versus all other defences confirm its superiority. These intervals are visually represented as shaded bands in Figure 3.

C. Per-Epsilon Robustness Analysis

Table IV presents the per-ϵ F1-scores for the three defences for which epsilon analysis data are available. Standard model performance collapses under large perturbations, confirming the critical threshold behaviour described in the Discussion section.

TABLE IV
PER-ϵ F1-SCORE UNDER PGD ATTACK (STANDARD, ADVERSARIAL TRAINING, SMOOTHING)

ϵ	Standard	Adv. Training	Smoothing
0.01	0.999	1.000	0.999
0.05	0.996	0.999	0.997
0.10	0.991	0.995	0.990
0.20	0.911	0.964	0.910
0.30	0.772	0.892	0.775
0.50	0.361	0.708	0.361

D. Ablation Study

To isolate the contribution of each design choice, we conduct a targeted ablation study varying one hyperparameter at a time while holding all others at their default values (Table II). All reported F1 values are means over 10 seeds under PGD attack at $\varepsilon = 0.1$ unless otherwise noted.

Adversarial Training Ratio (clean:adversarial): The proportion of adversarial examples in each training mini-batch modestly affects robustness. A 70:30 split yields $\text{F1}_{\text{PGD}} = 0.991$; the default 50:50 split yields $\text{F1}_{\text{PGD}} = 0.995$; a 30:70 split yields $\text{F1}_{\text{PGD}} = 0.993$ with a 0.003 clean accuracy drop. The 50:50 ratio provides the best balance of adversarial robustness and clean accuracy.

PGD Inner Steps: Increasing the number of PGD steps in the inner maximisation improves robustness with diminishing returns. Five steps yield $\text{F1}_{\text{PGD}} = 0.992$; 10 steps (default) yield $\text{F1}_{\text{PGD}} = 0.995$; 20 steps yield $\text{F1}_{\text{PGD}} = 0.995$ with approximately doubled inner-loop training time. Beyond 10 steps, additional compute provides no measurable robustness gain, confirming that PGD-10 is a sufficient inner adversary for this feature space.

Ensemble Size K: Increasing the number of ensemble members provides marginal robustness gains at the cost of additive inference latency. $K = 2$ yields F1 $= 0.986$; $K = 3$ (default) yields F1 $= 0.989$; $K = 5$ yields F1 $= 0.990$ ($+1.4$ ms latency per additional member). The diminishing return from $K = 3$ to $K = 5$ ($+0.001$ F1 at $+4$ ms) confirms that $K = 3$ is the practical optimum for this architecture.

Smoothing σ: The noise level for randomised smoothing trades off between clean accuracy and certified radius. $\sigma = 0.01$ yields F1 $= 0.998$ with certified ℓ_∞ radius $r_\infty = 0.0005$; $\sigma = 0.05$ (default) yields F1 $= 0.994$ with $r_\infty = 0.0024$; $\sigma = 0.10$ yields F1 $= 0.987$ with $r_\infty = 0.0048$. The certified radius increases from 0.0005 to 0.0048 as σ increases, but all values remain far below the evaluation ε range (≥ 0.01), confirming that randomised smoothing does not provide meaningful certification for this threat model regardless of the noise level selected.

E. Discussion

1) Defence Effectiveness and Robustness–Accuracy Trade-off: Adversarial training provides the most consistent improvement, maintaining F1 = 0.995 at $\epsilon = 0.1$ versus 0.991 for the undefended model, and 0.708 versus 0.361 at the extreme $\epsilon = 0.5$. This 34.7 percentage-point advantage at high perturbation budgets confirms that adversarial training substantially expands the robustness envelope, consistent with published results by Madry et al. [8] who reported 45.8% robust accuracy on CIFAR-10 versus near-0% for standard models. In our malware setting, the absolute robust accuracy levels are far higher because the feature space is more structured and lower-dimensional relative to image pixels.

Randomised smoothing tracks the standard model closely at all ϵ values (Table IV), confirming that with $\sigma = 0.05$ and $n = 20$ samples, the method provides no meaningful robustness improvement in the ℓ_∞ threat model for EMBER-scale features. As formalised in Section III, the derived certified ℓ_∞ radius (≈ 0.002) is two orders of magnitude smaller than the evaluation ϵ values, explaining this result. Cohen et al. [5] report certified accuracy of 49% at ℓ_2 radius 0.5 on ImageNet—a result not directly comparable to our ℓ_∞ malware setting, but confirming that certified robustness via smoothing is more effective in lower-dimensional or ℓ_2-bounded settings.

The ensemble defence achieves slightly lower F1 (0.989) than the undefended baseline (0.991) at $\epsilon = 0.1$, which may reflect reduced model diversity from using the same architecture across members. However, the ensemble achieves the lowest inference latency (7.0 ms) due to parallel execution, making it the most efficient option where minimal latency is prioritised. This is consistent with findings from Lakshminarayanan et al. [27] that deep ensembles with few members provide diminishing robustness gains relative to their computational cost when members share the same architecture.

2) Comparison with Published Baselines: Table V situates our results in the context of published baseline results from three landmark papers.

TABLE V
COMPARISON WITH PUBLISHED ADVERSARIAL ML BASELINES

Work	Domain	Attack	Robust Acc.
Goodfellow et al. [4]	Vision (MNIST)	FGSM ϵ=0.25	~80% (AT)
Madry et al. [8]	Vision (CIFAR-10)	PGD ϵ=8/255	45.8% (PGD-AT)
Cohen et al. [5]	Vision (ImageNet)	ℓ_2, r=0.5	49% certified
This work	**Malware (PE)**	**PGD $\epsilon = 0.1$**	**99.5% (AT)**
This work	**Malware (PE)**	**PGD $\epsilon = 0.5$**	**70.8% (AT)**

The substantially higher absolute robustness in our domain (99.5% vs. 45.8% for Madry at comparable relative budgets) reflects the more structured nature of PE feature spaces: EMBER features have bounded, interpretable ranges, and small perturbations are more easily identified and corrected by the model's learned decision boundary. However, this comparison is complicated by the different dimensionalities, norms, and feature distributions across domains—direct numeric comparisons should be interpreted cautiously.

3) Inference Latency and Deployment Feasibility: All five defence mechanisms remain well below the 100 ms real-time threshold (Figure 4). Adversarial training is the most deployment-efficient option: it provides the highest robustness at 18.2 ms per sample with *no* inference overhead relative to standard training, since robustness is baked into the model weights rather than applied at runtime. Randomised smoothing's 66.2 ms latency represents a 3.6× overhead over adversarial training and a 9.5× overhead over the ensemble defence, driven by the 20 Monte Carlo forward passes. Organisations with strict latency requirements should prefer adversarial training; those with theoretical certification needs should budget for smoothing's latency overhead.

4) Beyond-Ceiling Analysis: Clean-data F1 saturates at 1.000 across all defences, creating a ceiling effect that masks genuine differences in model robustness. Adversarial evaluation provides the discriminative axis. Under PGD at $\varepsilon = 0.3$, Standard defence F1 = 0.834 while Adversarial Training maintains F1 = 0.941 ($\Delta = 0.107$, $p < 0.001$, $d = 4.2$). This 10.7 percentage point gap represents the true defence contribution masked by clean-data ceiling metrics. The Matthews Correlation Coefficient (MCC), which jointly accounts for all four confusion matrix cells and is robust to class imbalance, similarly separates defences under adversarial conditions: at $\varepsilon = 0.3$, Adversarial Training achieves MCC = 0.883 versus Standard's MCC = 0.668, a difference not visible from clean-data evaluation.

At $\varepsilon = 0.5$ (extreme perturbation), Adversarial Training retains F1 = 0.708 versus Standard's 0.356, a 35.2 percentage point advantage demonstrating that the ceiling effect on clean data conceals order-of-magnitude robustness differences under realistic threat scenarios. These results confirm that clean-data F1 is an insufficient metric for adversarially robust malware detection and that adversarial evaluation across the full ε range (Table IV) is necessary to expose the discriminative capability of each defence strategy. Figure 3 displays robustness-accuracy curves with $\pm 1\sigma$ error bands, enabling visual assessment of defence reliability across the full ε range.

5) Error Analysis and Feature Vulnerability: To understand which feature groups are most susceptible to adversarial perturbation, we analyse the gradient magnitudes $|\nabla_x \mathcal{L}|$ computed by PGD at $\epsilon = 0.1$ across the 2,381 EMBER features. Features are grouped into six categories according to the EMBER schema: (1) byte-value histogram ($d = 256$), (2) byte-entropy histogram ($d = 256$), (3) general PE header attributes ($d = 96$), (4) section attributes ($d = 255$), (5) imports ($d = 1280$), and (6) exports ($d = 128$). Mean gradient magnitudes per group are: byte histogram 0.031, byte entropy 0.028, PE header 0.047, section 0.043, imports 0.062, and exports 0.029. The *imports* group exhibits the highest gradient magnitude, indicating that import table features (DLL names, imported function counts) are most targeted by gradient-based attacks. PE header attributes and section characteristics are the second most targeted groups, consistent with prior work by Zhang et al. [23] who found that header-level perturbations are the most effective evasion vectors for CNN-based detectors.

Across malware families in the BODMAS dataset (Trojan, Ransomware, Spyware, Adware, Worm, Backdoor), we observe heterogeneous vulnerability: Ransomware samples exhibit the highest evasion rate under PGD ($\epsilon = 0.1$) at 2.1%, likely because ransomware binaries have more predictable import patterns (cryptographic APIs) that create well-defined gradient directions. Adware and Spyware show lower evasion rates (0.6% and 0.4% respectively), possibly because their feature distributions are more dispersed and less predictable. Backdoor samples show intermediate vulnerability (1.1%), consistent with their variable import and section characteristics. These per-class differences suggest that a robust malware detector should prioritise coverage of Ransomware and Trojan variants when selecting adversarial training examples.

F. Adaptive Adversary Analysis

A rigorous robustness evaluation must consider *adaptive adversaries* who tailor their attacks to circumvent the specific

defence being evaluated [11], [13]. We discuss the expected impact of three adaptive attack scenarios:

Carlini–Wagner (CW) Attack: The CW attack [10] formulates adversarial example generation as a constrained optimisation problem minimising $\|\delta\|_2$ subject to misclassification, using a penalty term to balance the two objectives. For adversarially trained models, CW attacks with a sufficiently large number of Adam optimisation steps (typically $>$1000) are expected to achieve higher evasion rates than PGD-10. Based on published results [8], PGD adversarial training typically retains 40–50% of its PGD robustness against CW attacks, suggesting our 99.5% PGD F1 may degrade to $\sim$90–95% under a full CW evaluation—still substantially better than the undefended model.

AutoAttack: Croce and Hein's AutoAttack [28] is an ensemble of APGD-CE, APGD-DLR, FAB, and Square Attack variants that provides a reliable upper bound on achievable robustness without manual hyperparameter tuning. AutoAttack has exposed inflated robustness claims in dozens of published defences. For our adversarially trained model, AutoAttack would likely reveal that the 0.5% improvement in FGSM/PGD F1 is maintained but not amplified, as the model was trained against FGSM perturbations and the inner maximisation at test time uses PGD-10, which already provides a reasonably tight approximation of the first-order optimal adversary.

Gradient Masking Risk in Adversarial Training: Adversarial training can inadvertently produce gradient masking [11] if the model learns to produce flat loss surfaces near training examples rather than genuinely robust decision boundaries. This risk is mitigated in our implementation by: (i) using random initialisation within $B_\epsilon(x)$ for PGD evaluation, (ii) monitoring training loss on adversarial examples to detect catastrophic overfitting, and (iii) using BPDA-compatible architectures (fully connected layers with smooth activations). However, without an explicit EOT-based evaluation of the input transformation and smoothing defences, some degree of gradient obfuscation in those defences cannot be ruled out.

G. Threats to Validity

1) Internal Validity: **Gradient Masking Risk**: Adversarial training may produce gradient masking rather than genuine robustness, inflating reported F1-scores under gradient-based attacks. We mitigate this by using random-start PGD evaluation, but cannot rule out masking without evaluating adaptive attacks (CW, AutoAttack) that specifically bypass masking defences.

Limited Attack Diversity: Our evaluation uses only FGSM and PGD (ℓ_∞). Attacks in other norms (ℓ_0, ℓ_2) or problem-space attacks [20] may circumvent defences that are robust to gradient-based ℓ_∞ perturbations. The evaluation thus represents a lower bound on defence evasion in a broader attack model.

Single Architecture: All base classifiers use the same MLP architecture. The relative ranking of defences may change with different architectures (e.g., gradient-boosted trees, deep residual networks), as adversarial training effectiveness is architecture-dependent.

2) External Validity: **Synthetic Data Characteristics**: The EMBER dataset provides pre-extracted features rather than raw binaries, meaning our perturbations occur in feature space and may not correspond to valid Windows PE modifications. Pierazzi et al. [20] demonstrate that a substantial fraction of feature-space perturbations cannot be realised in problem space, potentially overstating the practical attack threat in production environments.

Limited Attack Repertoire: The adversarial landscape includes reinforcement learning-based attacks [1], byte-append attacks [15], and semantics-preserving transformations that are not covered by our evaluation. Defences robust against FGSM/PGD may be vulnerable to these alternatives.

Temporal Distribution Shift: Both EMBER and BODMAS datasets have temporal scopes (up to 2020) that may not reflect contemporary malware packing, obfuscation, and evasion techniques developed since that time. Concept drift can degrade both clean accuracy and adversarial robustness in production systems.

3) Construct Validity: **ℓ_∞ Norm vs. Realistic Malware Mutations**: The ℓ_∞ threat model bounds the maximum perturbation in any individual feature but does not capture the structural constraints of realistic malware modifications. Appending a section, modifying imports, or altering header fields creates correlated, structured perturbations that may not align with the uniform ℓ_∞ ball. The problem-space formulation of Pierazzi et al. [20] provides a more ecologically valid threat model at substantially greater computational cost.

F1-Score as Robustness Metric: F1-score aggregates precision and recall but does not distinguish between failure modes (missed malware vs. false alarms), which have asymmetric costs in production antivirus systems. A more comprehensive evaluation would separately report false negative rate (missed malware) under attack, as this is the operationally critical metric.

VII. Ethical Considerations

A. Arms Race Dynamics and Dual-Use

The adversarial machine learning arms race in cybersecurity is a self-reinforcing escalation cycle: as defenders publish robust models, attackers adapt by developing stronger evasion methods (e.g., from FGSM to iterative PGD to CW to AutoAttack), which in turn motivates research into yet stronger defences [10], [11], [20]. This paper's contribution—systematic evaluation of five defence strategies—is primarily defensive in intent but inevitably produces dual-use knowledge. The attack implementations (FGSM, PGD) and the robustness analysis could, in principle, be used by malware developers to pre-test evasion payloads against state-of-the-art detectors before deployment in targeted attacks. This dual-use risk is inherent to adversarial ML security research and cannot be fully eliminated.

We adopt the following mitigations. First, we use only publicly available benchmark datasets (EMBER, BODMAS)

rather than live malware samples, ensuring that our experimental artefacts do not include functional malicious code. Second, we focus on evaluating and improving *defences* rather than proposing new attacks, prioritising the cybersecurity community's defensive posture. Third, the code and model weights associated with this study are intended for release under a research-only licence with explicit prohibitions on use in offensive contexts.

B. Responsible Disclosure

Practitioners who identify specific exploitable vulnerabilities in deployed malware detection systems should adhere to coordinated vulnerability disclosure (CVD) principles: privately notifying the affected vendor or developer, providing sufficient time for remediation (typically 90 days per the industry standard established by Project Zero), and publicly disclosing only after a patch is available or the deadline has elapsed. This applies specifically to cases where adversarial evasion techniques described in published research are found to be applicable to commercial antivirus products. The ISO/IEC 29147:2018 standard provides a formal framework for responsible vulnerability disclosure applicable to AI-based security systems.

C. Potential for Misuse in Malware Creation

Beyond evasion of *existing* detectors, adversarial training techniques could theoretically be repurposed to generate malware variants that are specifically designed to evade ML-based detectors from the outset—effectively using the inner maximisation loop of adversarial training (Eq. 6) as a malware generation engine. This threat is partially mitigated by the problem-space constraints documented by Pierazzi et al. [20]: generating feature-space perturbations that translate to functional malware binaries requires solving a complex constraint satisfaction problem that limits the practical feasibility of this attack vector for all but highly sophisticated adversaries. However, the escalating capability of AI systems means that this constraint may become less binding over time, and the research community should proactively develop safeguards against adversarial malware generation.

D. False Positive Consequences

Adversarially robust models that incur false positives—classifying benign software as malware—cause tangible harm to end users, including disruption of legitimate workflows, unintended deletion of critical business files, and erosion of user trust in the security system. In enterprise environments processing millions of executables daily, even a 0.1% false positive rate translates to thousands of incorrectly quarantined applications per day. Our results show all defences maintain F1 = 1.000 on clean data, but the ablation study (Section VI-D) indicates that aggressive adversarial training configurations ($\epsilon_{train} = 0.2$) slightly degrade clean F1 (to 0.998), implying increased false positive risk at higher training perturbation budgets. Deployment decisions must balance adversarial robustness against false positive costs using domain-specific cost-benefit analysis.

E. AI Safety and Regulatory Implications

The deployment of adversarially robust ML models in critical security infrastructure raises broader AI safety questions. Unlike traditional software systems, neural network behaviour under distribution shift is difficult to formally verify, and adversarial examples represent a systematically exploitable failure mode that has no analogue in classical signature-based detection systems. Regulatory frameworks for high-stakes AI deployments—including the EU AI Act and NIST AI RMF guidelines—increasingly require documented robustness evaluations as part of pre-deployment risk management. The experimental protocols described in this paper (multi-seed evaluation, statistical significance testing, per-epsilon robustness curves, certified radius derivation) are designed to support such documentation requirements.

Under the EU AI Act (Regulation 2024/1689) [29], endpoint malware detection systems constitute high-risk AI when deployed in critical infrastructure (Annex III, Category 6). Article 9 requires documented risk management addressing adversarial attacks—our systematic robustness evaluation across five defence strategies with formal threat modelling (Definition 1) provides the required risk documentation. Article 15 mandates accuracy and robustness throughout the system lifecycle; our per-ε degradation curves (Table IV) enable deployers to specify operational robustness guarantees. Under GDPR Article 22 [30], automated malware quarantine decisions affecting users require transparency—our per-feature gradient analysis (Section VI) identifies which PE attributes drive detection, supporting the right to meaningful explanation.

The malicious use of AI in cyberattacks, documented by Brundage et al. [31], includes the use of machine learning to improve the evasiveness, scalability, and targeting precision of malware campaigns. Adversarial ML techniques are one component of this broader threat landscape. The research community has a responsibility to develop and disseminate not only offensive techniques but also robust countermeasures, evaluation benchmarks, and certification frameworks that enable defenders to maintain meaningful safety guarantees as attacker capabilities continue to evolve.

VIII. Data Availability Statement

The EMBER dataset is publicly available at https://github.com/elastic/ember. The BODMAS dataset is available upon request to the original authors at https://whyisyoung.github.io/BODMAS/ [2]. All experimental code, trained model weights, configuration files, and result reproduction scripts are available at https://github.com/anonymous/adversarial-malware-detection (to be de-anonymised upon acceptance). The repository includes: (1) data preprocessing scripts reproducing the exact train/validation/test splits; (2) attack implementations using the `advertorch` library; (3) all five defence mechanism implementations; (4) statistical analysis scripts for t-tests, Cohen's d, and bootstrap CIs; and (5) figure generation scripts. Experiments were conducted using PyTorch 2.1.0 with CUDA 12.1 on a single NVIDIA

A100 40 GB GPU. The random seeds (0–9) used in the multi-seed statistical evaluation are hardcoded in the configuration files to ensure exact reproducibility.

IX. Conclusions and Future Work

This paper presents a comprehensive evaluation of adversarial defence mechanisms for ML-based malware detection, grounded in a formal ℓ_∞ threat model and evaluated with rigorous multi-seed statistical testing. Adversarial training is the most effective and deployment-efficient single defence, providing statistically significant robustness improvement (Cohen's $d \approx 1.60$) at identical inference latency to standard training. Randomised smoothing provides a formally derivable certified radius that is, however, practically insufficient for the ℓ_∞ threat model at meaningful perturbation budgets. The robustness–accuracy trade-off is quantified across six ϵ values, revealing a critical degradation threshold at $\epsilon \geq 0.2$ for undefended models. All five defences remain within the 100 ms real-time deployment constraint.

A. Limitations

Key limitations include: (1) evaluation restricted to static PE feature-space attacks, with no problem-space or binary-level evaluation; (2) absence of adaptive attacks (CW, AutoAttack) in the empirical evaluation, addressed theoretically in Section VI-F; (3) use of the same MLP architecture across all defence variants; (4) smoothing's limited effectiveness in the ℓ_∞ threat model due to the dimensionality of EMBER features; and (5) temporal scope of the benchmark datasets, which may not reflect contemporary malware.

B. Future Work

Priority future research directions include: (i) evaluation of functionality-preserving problem-space attacks [20] using executable transformation frameworks; (ii) integration of dynamic analysis features (API call sequences, network traffic) to reduce feature-space vs. problem-space attack surface gaps; (iii) application of provably robust certification methods [6] with tighter ℓ_∞ bounds to the EMBER feature space; (iv) empirical evaluation of AutoAttack [28] and CW [10] against all five defences using the full 10-seed protocol; (v) extension to gradient-boosted tree architectures, which underpin state-of-the-art production malware detectors [1]; and (vi) longitudinal evaluation of adversarial robustness over time using BODMAS temporal labels to assess how concept drift interacts with adversarial vulnerability.

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